



Synoptic Project 2025

Introduction Presentation

Sally Earl

s.earl@uea.ac.uk

What is the Synoptic Project?

Group software development project (6 people)

Working on a live industry brief (Engineers without Borders)

Tying together all your learning from your time in CMP so far

Showcasing programming, systems development, and professional skills

Final outputs are assessed by markers at a group presentation.

How does it work?

We will provide a brief from Engineers Without Borders

This contains pages of information about an area of the world

You and your group will need to come up with a solution to a problem the people are experiencing



ENGINEERING FOR PEOPLE DESIGN CHALLENGE 2024-25

Makers Valley
Johannesburg, South Africa



Engineering for People Design Challenge

Makers Valley

The Area

Makers Valley is an informal term given to a collection of city suburbs on the eastern edge of inner Johannesburg, spanning from New Doornfontein to Bezuidenhout Valley, and including Bertrams, Judith's Paarl, Troyeville, and Lorentzville. Home to some 46,000 people, the historically economically disadvantaged area has rich cultural diversity and an evolving community culture but has also suffered from urban decay and high unemployment rates.

Population: Over 46,000 people*

Approximately 60% of Makers Valley residents were born outside of South Africa

44% of the population is aged between 15 and 34 years old.



Street art in Makers Valley

Makers Valley is now showing signs of regeneration, as activity from social and creative enterprises increases. This has led the neighbourhood to be informally defined by a group of entrepreneurs as Makers Valley, which reflects the presence of artists, craftspeople, urban cultivators, artisans, poets, designers, and youth advocates that live and work in the area. The valley has formed an identity rooted in creativity, and creative place making, representing a hub for social and creative enterprise.

The 'maker' movement challenges consumerism, with a do-it-yourself mentality, using traditional and modern methods, and encompasses not only artists and manufacturers, but anyone who wants to create something, be it dancers, poets, or gardeners. The idea is to empower people to explore their creativity whilst fostering a sense of community and belonging. The value is stemmed from the thought and attention to detail given to the product made, which is not likely to be mass produced, but rather carefully designed.

Watch this video that introduces Makers Valley, the place and the partnerships.

*As of 2020. Note this number may be higher due to a general population increase since and may also not account for the dynamic and often rapid changes in migration patterns.

Engineering for People Design Challenge

Overview

Johannesburg is a big, sprawling city with a transport system that is not always reliable and does not yet effectively connect to all the key areas of the city.

To connect with other parts of the city for work, study, or accessing other amenities, people make use of minibuses and public buses. Public bus networks commonly used by commuters in Johannesburg include the Metro Bus and Rea Vaya (meaning 'we are going'), both of which have been in operation for over 18 years. Car ownership is uncommon due to the high costs associated with purchasing and maintaining vehicles. Local car dealerships offer both new and used cars for sale. The affordability of minibus taxis has also decreased in recent years due to a significant rise in price. E-hailing services have become more affordable in recent years. They also offer provision for transport at night when minibus taxis are not operational.

Johannesburg has two rail networks: the MetroRail and the Gautrain. The MetroRail is a major rail line that runs through the city. The Gautrain is a rapid rail line that connects the city to the international airport. The Gautrain is a popular option for many commuters, particularly those who live in Makers Valley and need to travel to the airport.

Current Situation

In Makers Valley, the streets are generally wide. They often feature a pedestrian walkway and trees on either side of the tarmac road. Traffic lights (or 'robots' as they are known) are common at major junctions. Mini-buses are used to pick up children to and from school. Makers Valley residents often run these businesses alongside one or two others, such as owning a shop or working another job. There is a strong support network between these independent business owners. They are also among the few public transport options that offer flexible payment plans, including negotiated late payments for customers, particularly for the parents of schoolchildren.

Safety

Safety is a concern for pedestrians in Makers Valley. While risks persist both day and night — with instances of daytime muggings occurring in quieter areas — it is a concern for all. The occurrence of safety concerns is a concern for all.

Engineering for People Design Challenge



Chidi Maponya and Innocent Jiyane
Makers Valley Social Employment Fund

About

Innocent Jiyane, a program administrator for Makers Valley Partnership, oversees the SEF (Social Employment Fund), a government project aimed at employing members of the community, designed to tackle unemployment through the unique concept of social employment. The SEF participants work to maintain the community's health, cleanliness, and environmental sustainability. Chidi Maponya, as an operations coordinator, focuses on ensuring the smooth functioning of day-to-day activities.

In their roles, both Chidi and Innocent experience a range of challenges and deal with various people and situations, requiring strong managerial skills to address complaints and resolve issues. They find fulfilment in their work, particularly when they're able to employ individuals within the community, providing opportunities for them to contribute to keeping their surroundings clean and improving their living conditions.

Sharpen your problem solving skills because that's what we need right now. Problem solving skills for a better economy, lifestyle and wellbeing.

In their engagement with Makers Valley Partnership (MVP), they implement programs aimed at empowering community members. Chidi oversees a free digital literacy program tailored for community members, where individuals sign up to learn basic computer skills.

SEF aims to empower civil society organisations by enabling them to address issues in health, education, nutrition, gender-based violence and food security. MVP is one of the organisations benefiting from SEF, providing temporary work opportunities that contribute to the well-being and sustainability of the community.

Issues and Ideas

Challenges within the Makers Valley community include a lack of utilities. For example, access to the internet is limited and not available to everyone. Free internet access is only available from 9am to 4pm, leaving individuals disconnected from the online world outside of those hours. Chidi expresses a desire for greater access to internet and communication services for all community members.

Despite efforts by Maker's Valley to address crime through community involvement, safety concerns persist. This fear sometimes leads people to hesitate carrying necessary devices, like phones, which they rely on but fear could make them targets for theft. As a relatively small organisation, they face challenges in providing access to items like computers and cell phones to the community for free. What measures can be taken to ensure better access to the internet and computers in Makers Valley?

Watch Chidi and Innocent talk about their day to day experience in Makers Valley.

How does it work?

You will create designs, and then program a prototype over the course of 3 weeks

In the lab sessions, our TAs will support you with every step of the process

On the 12th June you will present your designs and prototype

Meet the Team

Module Organiser: Sally Earl (
s.earl@uea.ac.uk)

Additional Support:

Ellie Leist

Debbie Taylor

Hassan Malik

Raheel Hassan

Daniel Paredes-Soto

Raheel Hassan

Lab Support:

Catherine Ives-Keeler

Darcey Gardiner

Georgina Jeffries

Harry Whitelam

Andrew Driver

Chloe Evans

Alex Gwilliam

Charlotte Harris

Amber Yucebas

Ben Stannard

What is the Digital Design Challenge?

The Engineers without borders digital design challenge is a competition aimed at Engineering and Computer Science students

The brief details the problems that people in the area are facing every day

This year's brief is based on Makers Valley, Johannesburg in South Africa

You need to **design** and **build** a prototype piece of software to solve a problem you have identified.

Things to consider

How does your software help these people?

Is it sustainable?
Can it be maintained by its users?

Are there infrastructure problems (intermittent power, devices)?

Are the equipment requirements unreasonable?

Different type of project

Unlike other modules, we aren't going to tell you what to make!

You should focus on creating something which fits for the region, based on your own research

Read the Engineers without borders brief (Digital Design Challenge)

You can find this on Blackboard

You can make a website, app, software that would connect to a hardware component.... anything!

Professionalism

You should treat this project like a mini-internship

Each lab session you will have a brief 'stand-up' with a TA.

You are expected to arrive promptly to labs, and conduct yourself with professionalism throughout the project

Labs

Labs are compulsory, and contribute to your overall grade

Each lab session is worth 2 marks – one for attending, and one for engaging

If you need to miss a lab, you need to fill in the Synoptic Absence Form

You should also inform your group

You will also need to complete work outside of the labs

Absences

The absence form can be found here:

<https://forms.office.com/e/JfdXQDcXXg>

If your absence is 'approved', you will not lose the attendance mark for that session.

If your absence is 'rejected' you will lose the attendance mark.

Synoptic Project Absence Request
Form - 2025



Other Events

Careers Central is also running an event on Friday

Y1: 10am, LT2

Y2: 11am, LT2

This will look at using the skills you gain in the project to help you gain employment

1 mark is available for attending this event. If you cannot attend you must fill in the synoptic absence form

Timetable

Week Commencing 26th May

	Monday	Tuesday	Wednesday	Thursday	Friday		
09:00		Overview Lecture All Students LT1 9:00-12:00					
09:30							
10:00					Y1 Careers Workshop 10:00-11:00 LT2		
10:30							
11:00				Lab 11:00-12:30 Groups 1-13	Lab 11:00-12:30 Groups 38-49	Y2 Careers Workshop 11:00-12:00 LT2	(Drop-In Support)
11:30							
12:00					Drop-In Support		
12:30							
13:00		First Years 13:00-15:00 Breakout Rooms Location TBC					
13:30			Lab 13:30-15:00 Groups 14-25	Lab 13:30-15:00 Groups 50-60			
14:00							
14:30							
15:00		Second Years 15:00-17:00 Breakout Rooms Location TBC	Lab 15:00-16:30 Groups 26-37	Lab 15:00-16:30 Groups 61-71			
15:30							
16:00							
16:30							

The official timetable is on the Blackboard organisation
Face to face labs are in Sci 2.37

Timetable

Week commencing 2nd June

	Monday	Tuesday	Wednesday	Thursday	Friday
09:30					
10:00					
10:30					
11:00	Lab	Lab	Lab	Lab	Drop-In Support
11:30	11:00-12:30	11:00-12:30	11:00-12:30	11:00-12:30	
12:00	Groups 1-13	Groups 38-49	Groups 1-13	Groups 38-49	
12:30					
13:00					
13:30	Lab	Lab	Lab	Lab	
14:00	13:30-15:00	13:30-15:00	13:30-15:00	13:30-15:00	
14:30	Groups 14-25	Groups 50-60	Groups 14-25	Groups 50-60	
15:00	Lab	Lab	Lab	Lab	
15:30	15:00-16:30	15:00-16:30	15:00-16:30	15:00-16:30	
16:00	Groups 26-37	Groups 61-71	Groups 26-37	Groups 61-71	
16:30					

Timetable

Week Commencing 9th June

	Monday	Tuesday	Wednesday	Thursday	Friday
09:30				Marking	
10:00					
10:30					
11:00	Lab	Lab	Drop-In Support		
11:30	11:00-12:30	11:00-12:30			
12:00	Groups 1-13	Groups 38-49			
12:30					
13:00					
13:30	Lab	Lab			
14:00	13:30-15:00	13:30-15:00			
14:30	Groups 14-25	Groups 50-60			
15:00	Lab	Lab			
15:30	15:00-16:30	15:00-16:30			
16:00	Groups 26-37	Groups 61-71			
16:30					
17:30					
18:30					

Assessment – 12th June

The specific assessment criteria is on Blackboard

The expectation and marking criteria of 1st years and 2nd years is different due to the different levels of experience

Assessment – 12th June

The assessment criteria is split into six categories

- Problem Identification
- Design
- Implementation
- Sustainability
- Software Testing
- Presentation Quality

Assessment - 12th June

The presentation timetable will be made available next week

Presentation attendance is mandatory – you won't get any marks for the project if you do not attend

During the presentation you'll need to show your design work and a live demonstration of your solution

Anything you want marks on must be shown in the presentation

Assessment – Your mark

In addition to being assessed on your work, you will be required to fill out evaluations of your team members.

You must each fill out the evaluation form by 11:59pm,
12th June

Submission

Your work will be marked based on what you present.

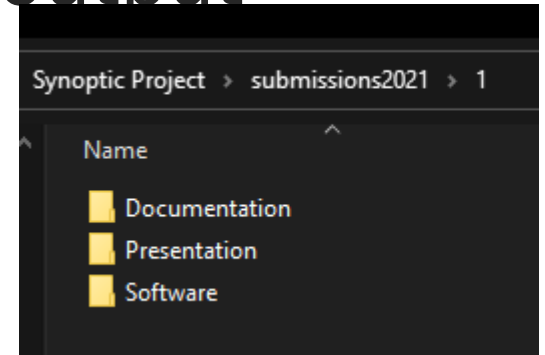
Group leader will be required to submit your output to Blackboard

This submission should contain:

- Your presentation

- Your software

- Any documentation you wish to show



Example
structure

What should I do if I have a problem?

In the first instance, bring any questions to the labs!

The drop-ins offer additional opportunities for help

Any questions outside of lab time should be emailed to me: s.earl@uea.ac.uk

Do not contact the TAs and other staff outside of the labs.

What you should do today

- Make a group chat on MS Teams
- Make a Trello board
- Find out each others skillsets
- Read the brief
- Attend the seminar sessions

First Years - 1pm-3pm

TEC 0.02	Groups 1-3
TEC 0.03	Groups 4-6
TEC 0.04	Groups 7-9
TEC 0.06	Groups 10-12
TEC 0.07	Groups 13-17
TEC 1.01	Groups 18-20
EH 1.25	Groups 21-25
EH 0.02	Groups 26-31
EH 0.12	Groups 32-37

Second Years - 3pm-5pm

TEC 0.02	Groups 38-40
TEC 0.03	Groups 41-43
TEC 0.04	Groups 44-46
TEC 0.06	Groups 47-49
TEC 0.07	Groups 50-54
EH 1.25	Groups 55-59
EH 0.02	Groups 60-65
EH 0.12	Groups 66-71



★ TEC: The Enterprise Centre

★ EH: Earlham Hall
0.02 and 1.25

★ 0.12